# H Sticking on Graphene Surface Including Vibrational Modes

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**Abstract:** The H sticking on graphene is to be investigated using Hybrid TRAJectory (HyTRAJ), a mixed quantum classical method. A single potential with well-defined barrier, physisorption and chemisorption well is to be obtained and the study of phonon vibrational modes using PWSCF method in Quantum Espresso.

## Introduction

**H Sticking Dynamics:** Understanding the sticking of H on graphene is important for low evergy intersteller medium where a physiosorbed H would be chemisorbed through quantum tunnelling.

- **Quantum Methods** (with T)
  - Morisset *et al.* [1] (DFT+CCWP method)
  - Karlicky *et al.* [2] (Chemisorption study using system bath model)
  - Lepetit *et al.* [3,4] (Physisorption study with several methods)
- **Mixed Quantum Classical** (with T)
  - Morisset *et al.* [5] (Classical Phonons + H motion 1D WP)

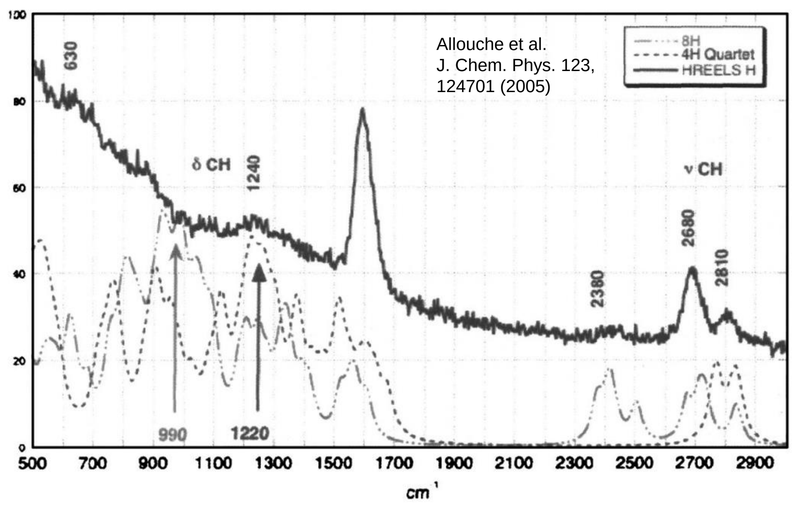
Challenges/Limitations
- Very Long
- One Phonon approximation
- Number of phonons at asymmetric points are limited
- Not very efficient to study molecules

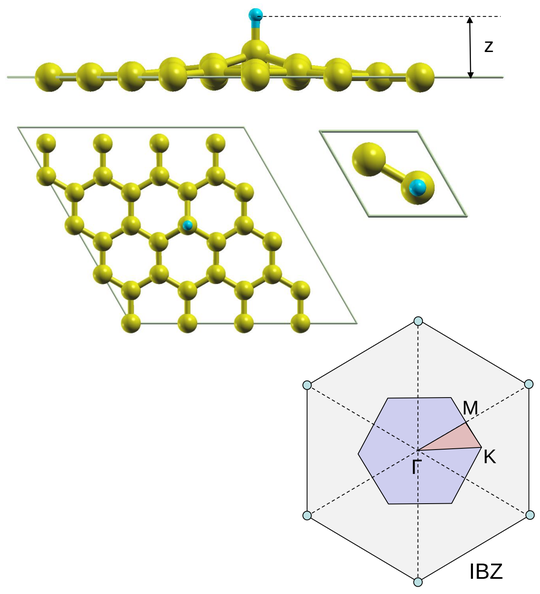
- **MCTDH**: Bonfanti *et al.* [6,7,8] (T=0K)

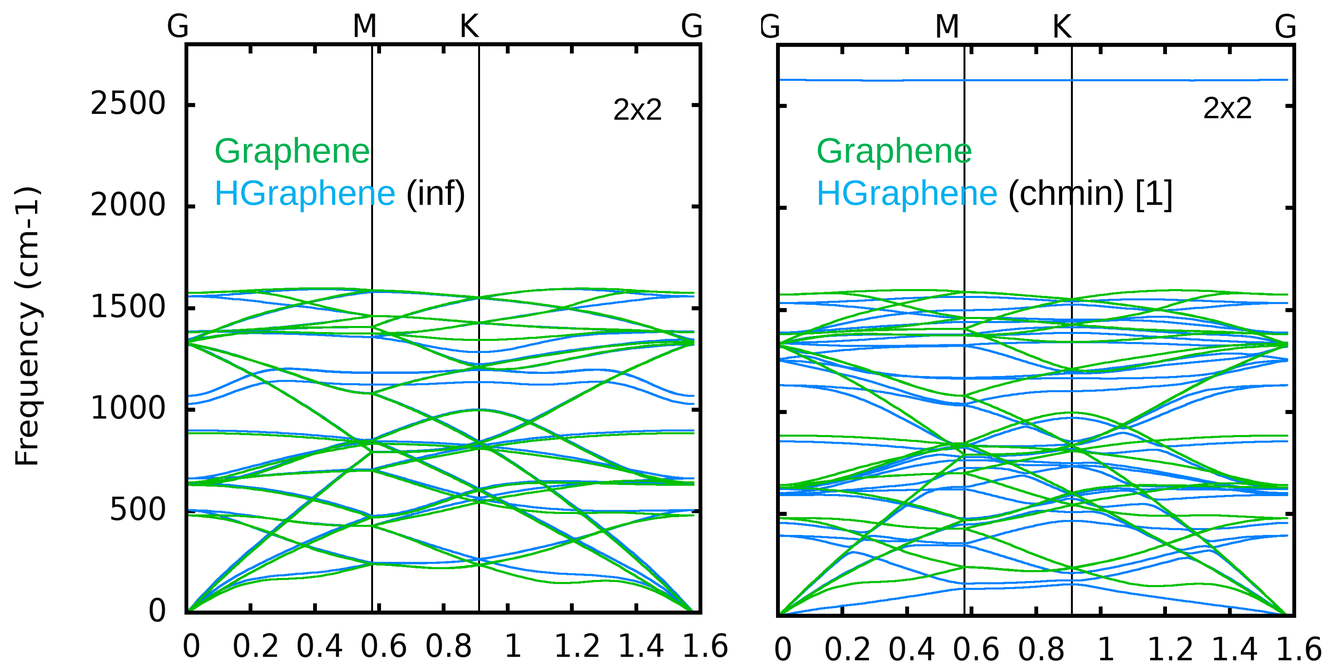
**Objective:** To study the sticking of H atom on graphenic surfaces using a new dynamics method developed in the ANR HyTRAJ (a hybrid quantum trajectories method) including phonon in dynamics.

Essential Ingredients: Single PEC, Phonon frequencies at Γ point and Phonon dispersion

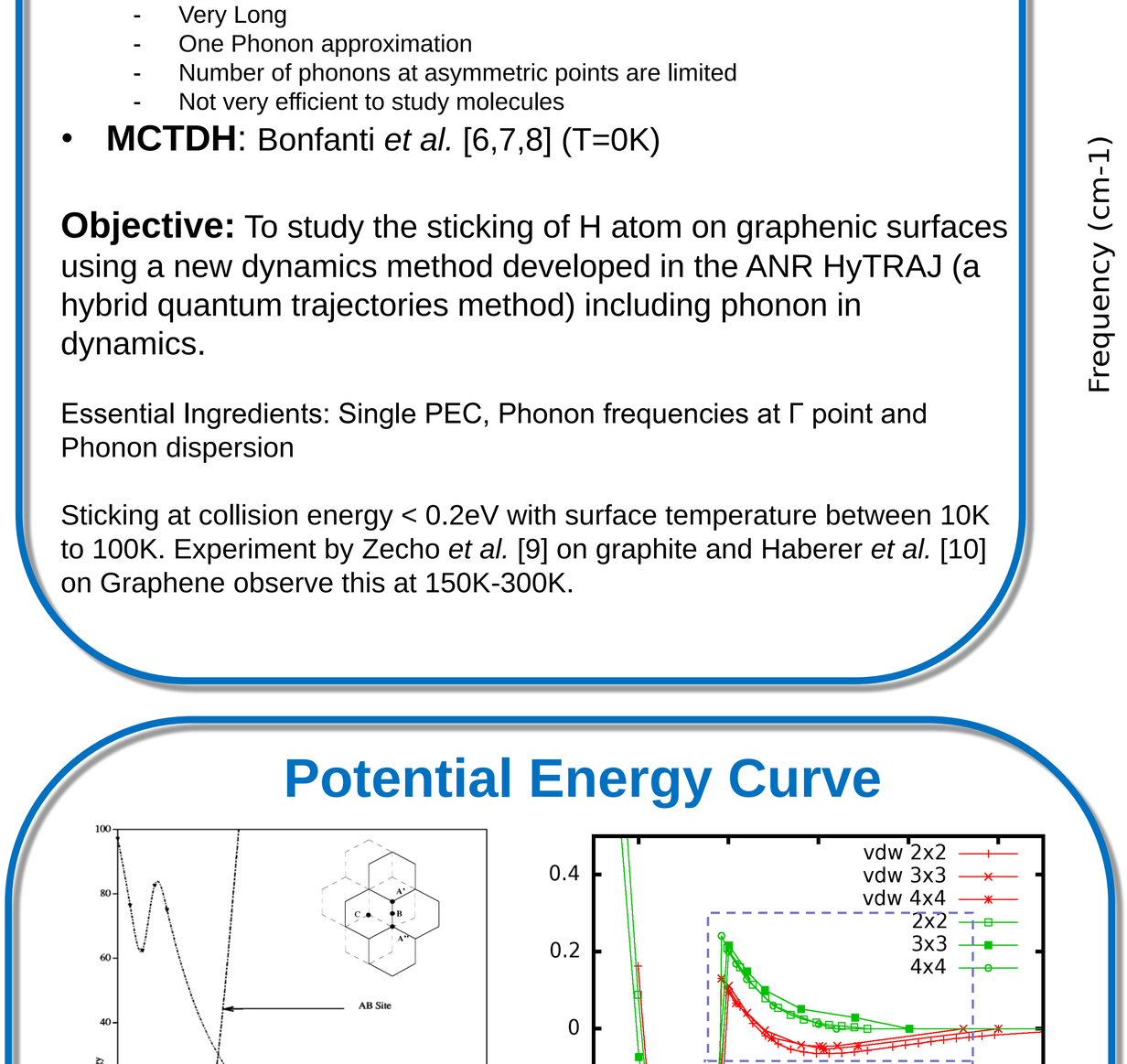
Sticking at collision energy < 0.2eV with surface temperature between 10K to 100K. Experiment by Zecho *et al.* [9] on graphite and Haberer *et al.* [10] on Graphene observe this at 150K-300K.




## Potential Energy Curve



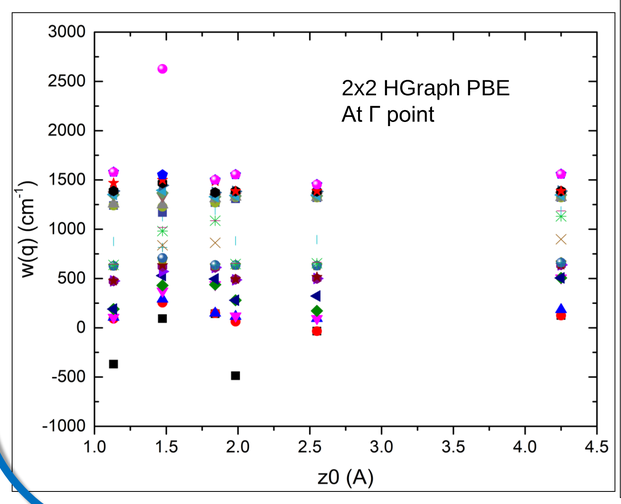

Ivanovskaya *et al.* / Rougeau *et al.* / Wang *et al.*

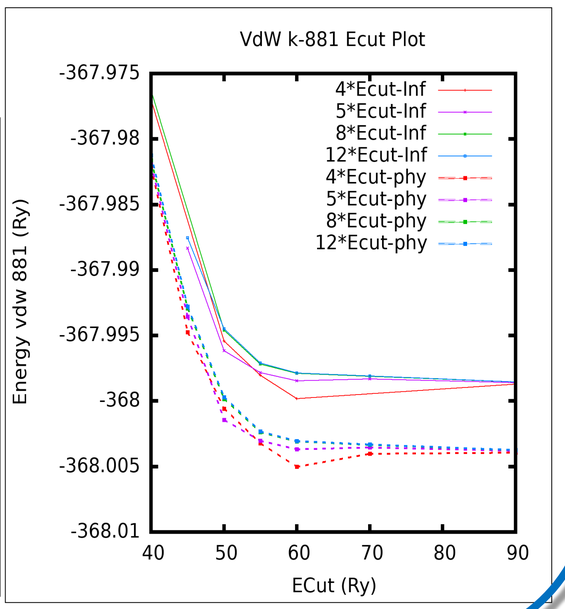
## Quantum Espresso

1. Solution for negative phonon frequencies ?
2. Calculation of ph.x in anharmonic approximation and its validity ?
3. Selection of Ecut/Erho ?

DFT (PBE) (GGA functionals) [16]
VdW-DFT (sla+pw+revx+vdw1) [17,18]

Projector Augmented-Wave (PAW) ultrasoft pseudopotentials (PPs) (rrkj3) [19]




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